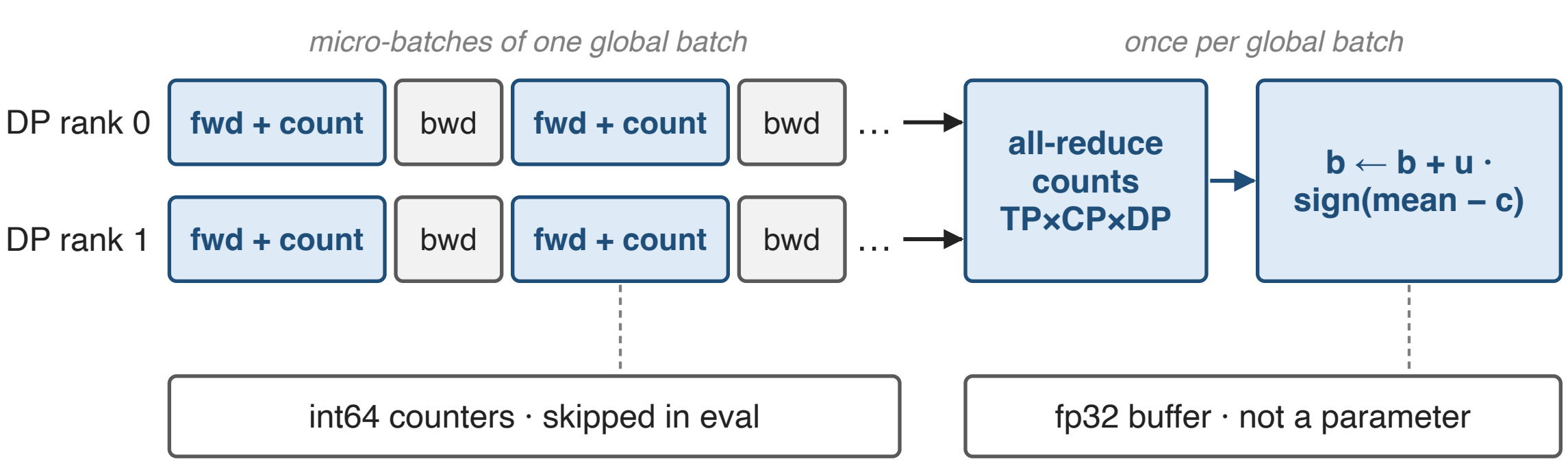
micro-batches of one global batch
once per global batch
DP rank 0
fwd + count
bwd
fwd + count
bwd
…
DP rank 1
fwd + count
bwd
fwd + count
bwd
…
all-reduce counts TP×CP×DP
b ← b + u · sign(mean – c)
int64 counters · skipped in eval
fp32 buffer · not a parameter